THE SIMPLE PENDULUM: Measurement of g

PURPOSE

The objective of this Laboratory Exercise is to measure the acceleration due to gravity in this room using the simple pendulum as a gravity sensor.

EQUIPMENT AND SUPPLIES

Pendulum bob and string, sturdy support, metric ruler, and timer.

INTRODUCTION

A pendulum is called simple if all the mass is concentrated at a point at the end of a massless string. We get a good approximation of a simple pendulum by using a heavy "bob" of mass m suspended by a light string. The pendulum length L is taken to be the distance from the support to the center of the pendulum bob.

$$T^2 = 4\pi^2 L/g$$

The period T of a simple pendulum of length L oscillating with small amplitude is given as where g is the acceleration due to gravity. Note that the mass of the bob does not appear in the equation. For our purposes, "small amplitude" means an angular displacement of less than 10°. Solving the above equation for g, we find

$$g = 4\pi^2 L/T^2$$

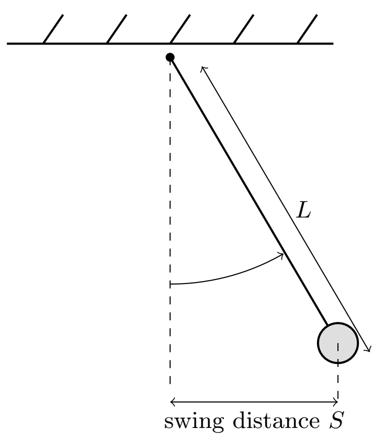

Figure 1: a simple pendulum

PROCEDURE

1. Set up the pendulum at its longest length, about 180cm. On the data sheet provided record your measured value of L. Note that L, the length of the pendulum, is the sum of the length of the string plus the length of the hook on the bob plus the radius of the bob.

For four different scenarios, measure the time required for 25-50 complete oscillations (oscillation is complete when the pendulum has returned to its starting point). For each scenario, repeat this measurement three times. Enter the number of oscillations, N, and the three times you measured on the data sheet.

2. At its longest length, set the pendulum swinging through approximately ± 10 cm.
3. With it still at its longest length, set the pendulum swinging through approximately ± 5 cm.
4. Shorten the pendulum to a medium length, about 150 cm. Set the pendulum swinging through approximately ± 5cm.
5. Shorten the pendulum to a short length, about 100 cm . Set the pendulum swinging through approximately ± 5cm.
6. For each pendulum length, compute the average period, T, for the oscillations. Then complete the remaining columns on data sheet for each pendulum length.

Calculation 1: Average the three determinations of g that you have now measured.

Calculation 2: To within what percent is your result in agreement with the accepted value of 980 cm/sec^2 ?

Percent Deviation

$$\text{Percent Deviation} = \left| \frac{\text{Your Value} - \text{Accepted Value}}{\text{Accepted Value}} \right| \times 100\%.$$

7. Plot the best-fit line, also called regression line, using spreadsheet software (for example, MS Excel or Google Sheets). Plot the values of $4\pi^2$ L on the vertical y-axis and the corresponding values of T^2 on the horizontal x-axis. Include the origin (0,0) in your plot. The regression line gives you the slope of the best-fit line.

Calculation 3: Figure out how to use this graph to estimate g. Compare the value of g deduced from the graph with the value of g you obtained in Step 6.

DATA SHEET

Name: _____________ Date: ___________

Instructor: _____________ Partners: ___________________________

Swing Distance $\boldsymbol{S}$ (cm)	Pendulum length $\boldsymbol{L}$ (cm)	Time for N oscillations (sec)	T Average period for 1 oscillation (sec)	$\boldsymbol{T^2}$ (sec^2)	$\boldsymbol{g} = \frac{4\boldsymbol{\pi}^2\boldsymbol{L}}{\boldsymbol{T}^2}$ (cm/sec^2)
$\pm$10 cm	long	N = 1) $t =$ 2) $t =$ 3) $t =$			
$\pm$ 5cm	long	N = 1) $t =$ 2) $t =$ 3) $t =$			
$\pm$5 cm	medium	N = 1) $t =$ 2) $t =$ 3) $t =$			
$\pm$5 cm	short	N = 1) $t =$ 2) $t =$ 3) $t =$			

Results

1. From your experimental results, compute the average value of g.
2. Compare the value of g you calculated above to the accepted value of $g = 9.80$ m/sec^2 $\left(980 \text{ cm/sec}^2\right)$. Express your answer as a percent deviation.
3. What is the value of g obtained from the slope of the graph from Step 7?

Aim

Results

Conclusion